

Mini Exercise 2 – Feature Engineering

1. The file **mini_data.csv** contains 3 features (X, Y and Z) and the table below shows our **entire** dataset.

X	Y	Z
1.3	1.0	1.2
3.5	3.4	3.3
3.2	2.9	3.1
1.1	0.9	1.1
1.4	1.2	1.1
0.9	0.8	0.9
4.2	3.9	4.1
3.8	3.9	4.2
1.2	1.0	1.0
3.7	3.5	3.7

2. The dataset has **not** been standardized.

3. Write Python code to perform **standardization** on the data, and **show** that your standardized data (i.e. each of your non-categorical features) has **zero-mean** and **unit-variance**. Feel free to use any NumPy functions, such as `numpy.mean()` and `numpy.std()`, in your solution.

4. Find out what happens if we perform standardization on a dataset that has already been standardized? Explain the outcome.